

Testing whether Q and R are properly set

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Outline

- 1 Model and Notation
- 2 Formal Assumptions and Propositions
- 3 Diagnostic Tests
- 4 NIS/NEES example plots
- 5 Summary

Linear Gaussian state-space model

Consider the discrete-time linear model given by

$$\begin{aligned}x_k &= F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + w_{k-1}, \\y_k &= H_k x_k + v_k,\end{aligned}$$

with

$$w_{k-1} \sim \mathcal{N}(0, Q_{k-1}), \quad v_k \sim \mathcal{N}(0, R_k).$$

- $x_k \in \mathbb{R}^n$: state
- $u_k \in \mathbb{R}^r$: known input
- $y_k \in \mathbb{R}^m$: measurement
- $Q_k \in \mathbb{R}^{n \times n}$: process noise covariance
- $R_k \in \mathbb{R}^{m \times m}$: observation noise covariance

Kalman notation

Prediction/update notation:

$$\hat{x}_{k|k-1}, P_{k|k-1} \quad \text{and} \quad \hat{x}_{k|k}, P_{k|k}.$$

Errors:

$$\tilde{x}_{k|k-1} = x_k - \hat{x}_{k|k-1},$$

$$\tilde{x}_{k|k} = x_k - \hat{x}_{k|k}.$$

Innovation and innovation covariance:

$$\nu_k = y_k - H_k \hat{x}_{k|k-1},$$

$$S_k = H_k P_{k|k-1} H_k^\top + R_k.$$

Whitened innovation:

$$e_k = S_k^{-1/2} \nu_k.$$

Kalman filter recursion

Prediction:

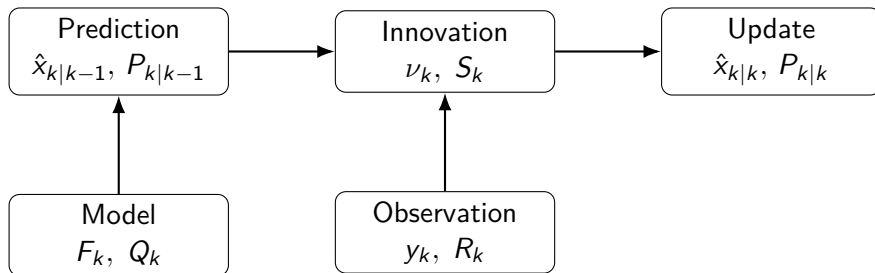
$$\begin{aligned}\hat{x}_{k|k-1} &= F_{k-1}\hat{x}_{k-1|k-1} + G_{k-1}u_{k-1}, \\ P_{k|k-1} &= F_{k-1}P_{k-1|k-1}F_{k-1}^\top + Q_{k-1}.\end{aligned}$$

Update:

$$\begin{aligned}K_k &= P_{k|k-1}H_k^\top S_k^{-1}, \\ \hat{x}_{k|k} &= \hat{x}_{k|k-1} + K_k\nu_k, \\ P_{k|k} &= (I - K_kH_k)P_{k|k-1}.\end{aligned}$$

Joseph form:

$$P_{k|k} = (I - K_kH_k)P_{k|k-1}(I - K_kH_k)^\top + K_kR_kK_k^\top.$$



Assumptions

Assumption 1 (Linear-Gaussian and independence)

For each k ,

- 1 the system is linear:

$$x_k = F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + w_{k-1}, \quad y_k = H_k x_k + v_k;$$

- 2 $w_k \sim \mathcal{N}(0, Q_k)$, $v_k \sim \mathcal{N}(0, R_k)$;
- 3 $\{w_k\}$ and $\{v_k\}$ are temporally white;
- 4 w_i and v_j are mutually independent for all i, j ;
- 5 the filter is initialized with the correct first and second moments.

Proposition: innovation covariance

Proposition 1

Under Assumption 1, the innovation

$$\nu_k = y_k - H_k \hat{x}_{k|k-1}$$

has mean zero and covariance

$$\text{Cov}(\nu_k) = S_k = H_k P_{k|k-1} H_k^\top + R_k.$$

Covariância mede o grau de interdependência entre duas variáveis aleatórias.

Autocovariância mede a persistência do sinal

Proof.

Using

$$\nu_k = H_k(x_k - \hat{x}_{k|k-1}) + v_k = H_k \tilde{x}_{k|k-1} + v_k,$$

we get $\mathbb{E}[\nu_k] = H_k \mathbb{E}[\tilde{x}_{k|k-1}] + \mathbb{E}[v_k] = 0$, and, by independence,

$$\text{Cov}(\nu_k) = H_k P_{k|k-1} H_k^\top + R_k.$$

Covariância também pode ser utilizada para analisar o lag de um sinal, ou seja, se os valores futuros podem ser relacionados com os valores anteriores. Se $\text{cov}(x, y) = 0$, o sinal é puro ruído. Se $\text{cov}(x, y)$ for alto, então o sinal é smooth (suave) na variação.

Proposition: whitened innovation properties

Proposition 2

Assume S_k is positive definite and define

Covariância da inovação

$$e_k = S_k^{-1/2} \nu_k.$$

Covariância da inovação

Inovação

Then

$$\mathbb{E}[e_k] = 0, \quad \text{Cov}(e_k) = I_m.$$

Moreover, under the ideal Kalman assumptions, $\{e_k\}$ is white.

Remark 1

This proposition explains why **whitened innovation diagnostics** are central for testing whether Q and R are plausibly set.

Isso retorna uma forma de analisar se os diagnósticos de inovação são plausíveis ou não.

Proposition: NIS distribution

Proposition 3 (Normalized Innovation Squared)

Define the NIS statistic

$$\epsilon_{z,k} = \nu_k^\top S_k^{-1} \nu_k = e_k^\top e_k.$$

$e_k = S_k^{-1/2} \nu_k$ (inovação)

Under Assumption 1,

$$\epsilon_{z,k} \sim \chi_m^2.$$

O artigo mostra que isso só é verdade quando o filtro está sintonizado.

Hence

$$\mathbb{E}[\epsilon_{z,k}] = m.$$

Proposition: NEES distribution

Proposition 4 (Normalized Estimation Error Squared)

Suppose the true state x_k is available and define

$$\epsilon_{x,k} = \tilde{x}_{k|k}^\top P_{k|k}^{-1} \tilde{x}_{k|k}.$$

Under Assumption 1,

$$\epsilon_{x,k} \sim \chi_n^2, \quad \mathbb{E}[\epsilon_{x,k}] = n.$$

Remark 2

NIS is a measurement-space consistency test; NEES is a state-space consistency test.

Corollary: averaged NIS and NEES

Corollary 1

For N samples, define

$$\bar{\epsilon}_z = \frac{1}{N} \sum_{k=1}^N \epsilon_{z,k}, \quad \bar{\epsilon}_x = \frac{1}{N} \sum_{k=1}^N \epsilon_{x,k}.$$

Then, under Assumption 1,

$$N\bar{\epsilon}_z \sim \chi_{Nm}^2, \quad N\bar{\epsilon}_x \sim \chi_{Nn}^2.$$

Thus level- α acceptance intervals are

$$\frac{1}{N} \chi_{\alpha/2, Nm}^2 \leq \bar{\epsilon}_z \leq \frac{1}{N} \chi_{1-\alpha/2, Nm}^2,$$

and

$$\frac{1}{N} \chi_{\alpha/2, Nn}^2 \leq \bar{\epsilon}_x \leq \frac{1}{N} \chi_{1-\alpha/2, Nn}^2.$$

Mathematical diagnostic protocol

Test the following:

① Innovation mean:

$$\hat{\mu}_\nu = \frac{1}{N} \sum_{k=1}^N \nu_k$$

should be near zero.

② Whitened innovation covariance:

$$\hat{\Sigma}_e = \frac{1}{N-1} \sum_{k=1}^N (e_k - \hat{\mu}_e)(e_k - \hat{\mu}_e)^\top \approx I.$$

③ Whiteness:

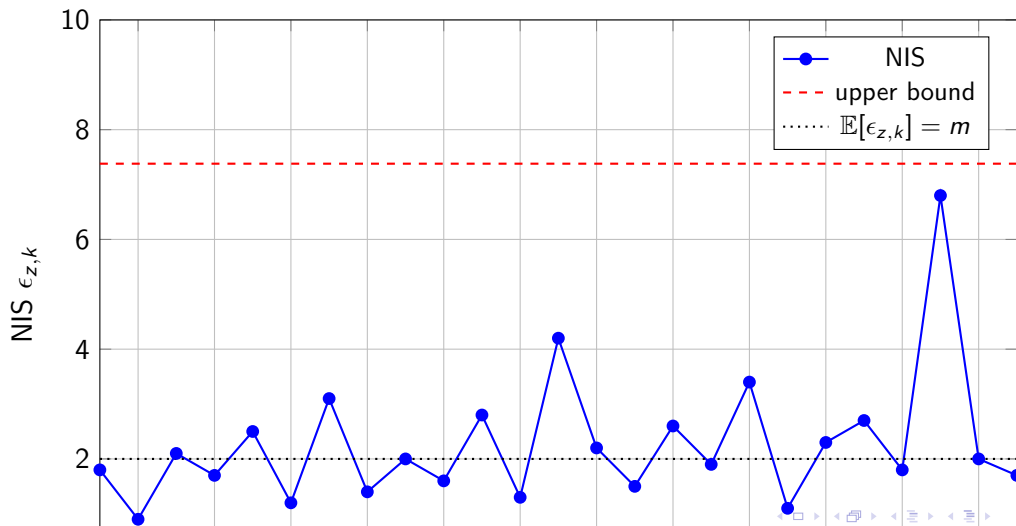
$$\hat{\Gamma}_e(\ell) = \frac{1}{N-\ell} \sum_{k=\ell+1}^N e_k e_{k-\ell}^\top \approx 0, \quad \ell \neq 0.$$

④ NIS/NEES consistency via chi-square bounds.

Observed result	Interpretation
$\bar{\epsilon}_z$ too high	innovation uncertainty underestimated; often R too small and/or Q too small
$\bar{\epsilon}_z$ too low	innovation uncertainty overestimated; often R too large and/or Q too large
$\bar{\epsilon}_x$ high, NIS acceptable	process side under-modeled; Q too small or dynamics wrong
$\hat{\mu}_e \neq 0$	bias or missing state
$\hat{\Gamma}_e(\ell) \neq 0$	colored noise, model mismatch, or poor Q structure

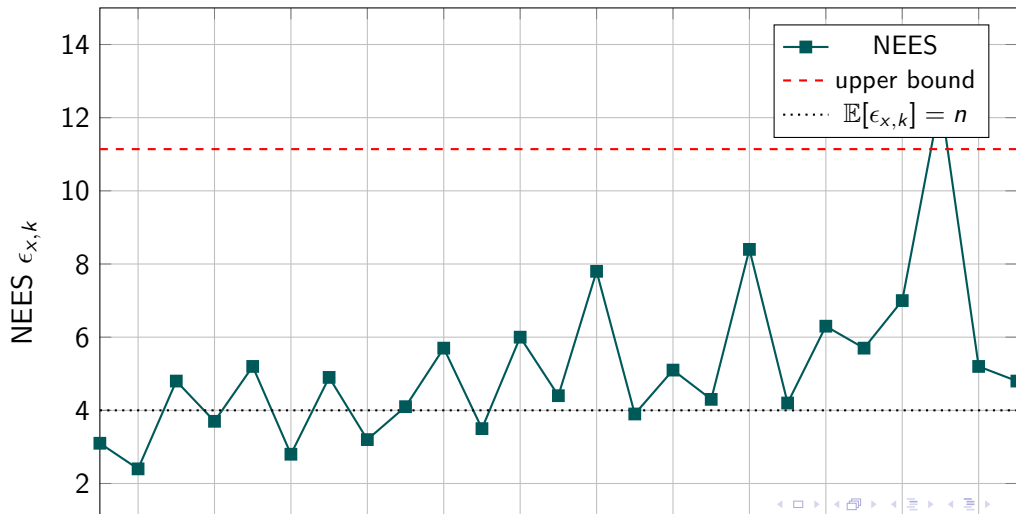
NIS sequence with pointwise confidence bounds

NIS sequence, $m = 2$, 95% pointwise bounds



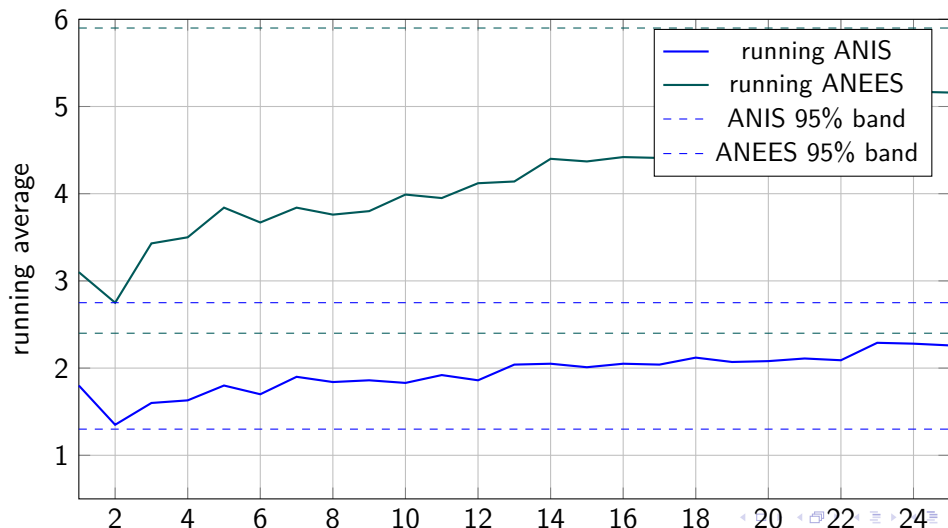
NEES sequence with pointwise confidence bounds

NEES sequence, $n = 4$, 95% pointwise bounds



Averaged NIS and NEES

Running averages of NIS and NEES



- The central objects are

$$\nu_k, \quad S_k, \quad e_k, \quad \epsilon_{z,k}, \quad \epsilon_{x,k}.$$

- Well-set (Q, R) should yield:
 - nearly white, zero-mean whitened innovations,
 - $\text{Cov}(e_k) \approx I$,
 - NIS and NEES consistent with chi-square theory.
- Failed diagnostics do not uniquely blame Q or R :
 - missing biases,
 - poor dynamics,
 - bad observation model,
 - colored disturbances

may be the real cause.

Does it make sense?

Any questions?